Homework: LLM and its Application

Q1. Compare BERT, GPT, and T5 in terms of architecture, pre-training objectives, and typical applications. Consider a decoder-only Transformer model with the following configuration:

- Number of layers, n=24
- Hidden dimension, d=2048
- Number of attention heads, h=16
- Feed-forward dimension, d_{ff}=8192

Assume that the approximate number of parameters per Transformer layer is: $P_{layer}=12d^2$

a) Calculate the approximate number of parameters in one Transformer layer.
b) Estimate the total number of parameters in all 24 Transformer layers.
c) If the vocabulary size is 50,000 and the embedding dimension is 2048, calculate the number of input embedding parameters.
d) Calculate the approximate total parameter count, including the embedding parameters.

Q2. What are emergent abilities in LLMs? Explain how scaling can lead to the emergence of new capabilities, with suitable examples. Consider an LLM is trained using two different datasets.

- Dataset A contains 100 billion tokens.
- Dataset B contains 400 billion tokens.

Assume that the training loss follows the simplified scaling relationship: $L(D) = \left(\frac{D_c}{D}\right)^{\alpha_D}$

where D is the dataset size, D_c is a constant, and $\alpha_D = 0.5$.

a) Calculate the ratio of the loss for Dataset B to the loss for Dataset A, assuming the same model and compute configuration.
b) If the loss for Dataset A is 2.0, estimate the loss for Dataset B.
c) Explain the significance of scaling laws in planning the training of large language models.

Q3. Explain the complete training pipeline of a modern LLM. Explain the major architectural modifications introduced in Llama 2 compared with the conventional Transformer architecture. Consider Llama 2 is available in three configurations: 7B, 13B, and 70B parameters. Assuming FP16 precision:

a) Calculate the memory required to store each model.
b) Calculate the additional memory required to store the 70B model compared with the 7B model.
c) If a GPU has 24 GB memory, which of these models can theoretically fit based on parameter storage alone?
d) Why is actual GPU memory consumption higher than the parameter storage requirement?

Q4. Explain the RLHF framework with a suitable block diagram. Also discuss why RLHF is important for aligning LLM outputs with human preferences. Consider an LLM generates three responses (A, B, and C) to the same user prompt. Human evaluators rank the responses as follows:

$$A > B > C$$

A trained reward model assigns the following scores:

$$r(A) = 2.5, \quad r(B) = 1.5, \quad r(C) = 0.5$$

The probability that response A is preferred over response B is calculated using the Bradley-Terry model: $P(A > B) = \frac{e^{r(A)}}{e^{r(A)} + e^{r(B)}}$

a) Calculate the probability that response A is preferred over response B.
b) Calculate the probability that response B is preferred over response C.
c) Calculate the probability that response A is preferred over response C.
d) Explain how a reward model learns human preferences from ranked responses.

Q5. What is Chain-of-Thought (CoT) prompting? Explain how breaking a complex problem into intermediate reasoning steps can improve performance on certain reasoning tasks. Consider the following problem:

> A company has 240 employees. Of these, 35% are working in the research department. Among the remaining employees, 40% are working in the development department. The rest are working in administration. Calculate the number of employees in each department.

a) Explain how the problem can be solved using Direct Prompting, where the LLM is asked to provide only the final answer.
b) Formulate a suitable Chain-of-Thought (CoT) prompt that encourages the LLM to break the problem into intermediate reasoning steps before producing the final answer.
c) Demonstrate the expected reasoning steps and calculate the number of employees in each department.
d) Compare Direct Prompting and CoT Prompting in terms of reasoning transparency, suitability for multi-step problems, and answer verification.

Q6. Explain the principle of Low-Rank Adaptation (LoRA) with a suitable mathematical formulation and diagram. State two advantages of LoRA over full fine-tuning. Consider a pre-trained weight matrix W of dimension: $W \in \mathbb{R}^{4096\times4096}$

The model is adapted using LoRA with rank r=16. LoRA represents the weight update as:

$$W' = W + \frac{\alpha}{r} BA$$

where: $A \in \mathbb{R}^{r\times4096}$, $B \in \mathbb{R}^{4096\times r}$

a) Calculate the number of parameters in the original weight matrix.
b) Calculate the total number of trainable parameters in matrices A and B.
c) Calculate the percentage reduction in trainable parameters using LoRA compared with full fine-tuning.
d) If alpha=32, calculate the scaling factor.

Q7. Define perplexity and explain how it measures the predictive performance of a language model. Consider two language models, Model A and Model B, are evaluated on the same test sequence of 100 tokens. The total negative log-likelihoods are:

- Model A: 120
- Model B: 90

a) Calculate the average cross-entropy loss for both models.
b) Calculate the perplexity of both models.
c) Identify which model performs better in terms of next-token prediction and justify your answer.

Q8. Explain the concept of multimodal LLMs and the need for cross-modal learning. Explain how models such as CLIP facilitate the integration of image and text representations. Consider the following two systems:

- System A: Accepts only text input and generates text responses.
- System B: Accepts text and image inputs and generates text responses.

A user provides the following input:

"What objects are present in this image, and what is the relationship between them?" along with an image containing a person riding a bicycle.

a) Explain how System A and System B would handle this query.
b) Compare the capabilities and limitations of the two systems.

Q9. Explain the major developments in GPT-3 compared with earlier GPT models. Why can GPT-3 perform certain tasks without task-specific fine-tuning? Explain why instruction tuning improves the ability of GPT models to follow user instructions.

Q10. Consider an organisation deploying an LLM for a customer-support chatbot. The system receives 1,000 user queries per hour and must generate accurate responses with minimum delay.

a) Identify four challenges that may arise while deploying the LLM for this application.
b) Explain the trade-off between response quality, inference latency, and computational cost.
c) Suggest three measures to improve the efficiency and reliability of the chatbot.